



ProbloU⁺

Enhanced Probabilistic IoU Loss for Oriented Object Detection

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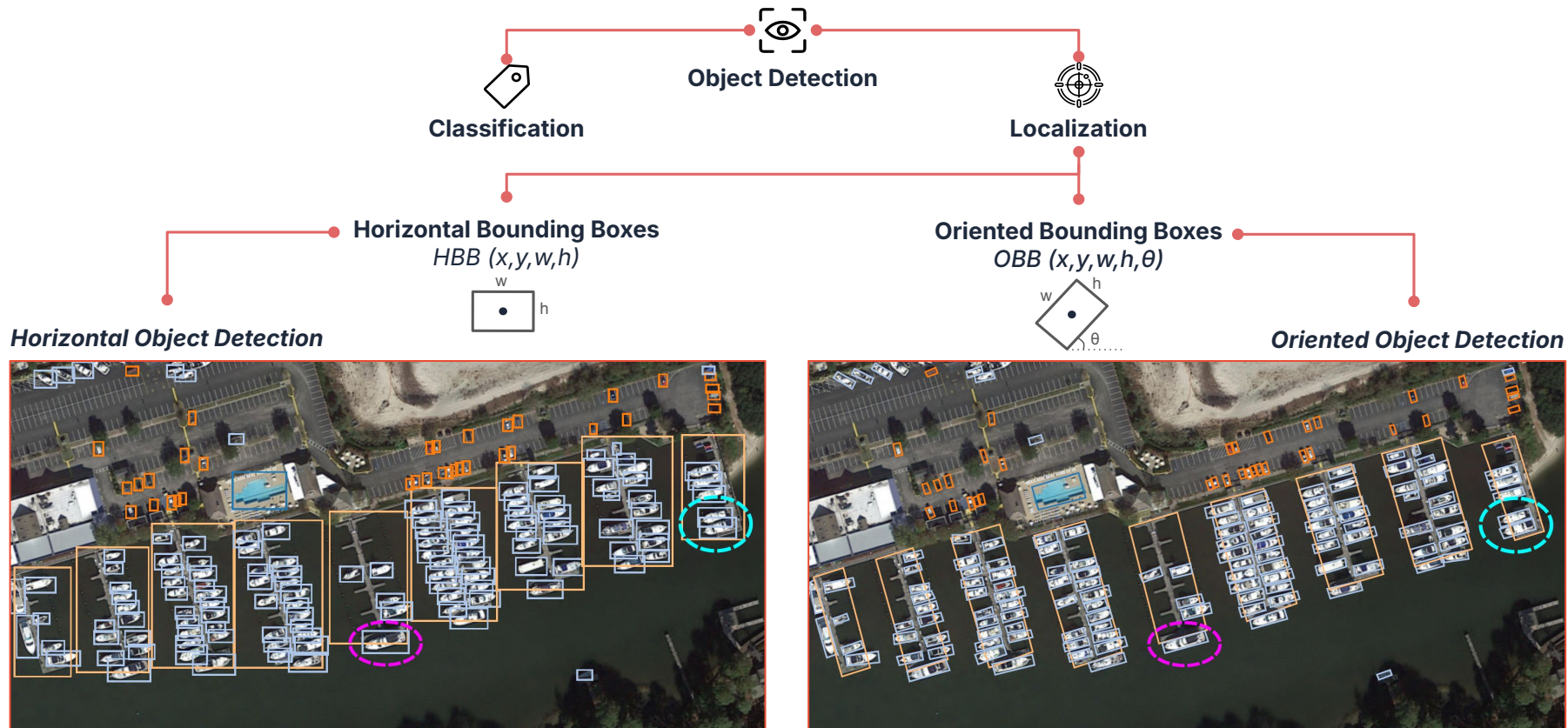
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4



Object Detection



Aerial image from the DOTA^{ref.1} dataset

Localization Losses for Horizontal Object Detection

Smooth L1 Loss^{ref.2}

The standard regression loss used in Fast-RCNN^{ref.2}.

😊 Provides a balanced trade-off between the L1 and L2 losses.

😞 Assumes parameter independence and lacks scale invariance.

$$L(z_{pred}, z_{gt}) = \begin{cases} 0.5(z_{pred} - z_{gt})^2 & \text{if } |z_{pred} - z_{gt}| < 1 \\ |z_{pred} - z_{gt}| - 0.5 & \text{otherwise} \end{cases}$$

$$Loss_{smooth\ L1} = \sum_{z \in \{x, y, w, h\}} L(z_{pred}, z_{gt})$$

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Intersection over Union (IoU) Loss^{ref.3}

Directly optimizes the geometric overlap between boxes.

- 😊 Scale-invariant.
- 😞 Zero gradient for non-overlapping boxes (IoU = 0).

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} = \frac{|B_{pred} \cap B_{gt}|}{|B_{pred} \cup B_{gt}|}$$

$$Loss_{IoU} = 1 - IoU$$

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Generalized IoU (GIoU) Loss^{ref.4}

👤 Addresses the vanishing gradients of IoU Loss.

🔍 Introduces a penalty based on the smallest enclosing box covering both boxes.

Distance IoU (DIoU) Loss^{ref.5}

👤 Addresses the slow convergence of GIoU Loss.

🔍 Introduces a penalty based on the distance between box centers.

Complete IoU (CIoU) Loss^{ref.5}

👤 Improves over DIoU Loss.

🔍 Adds a penalty to the DIoU based on the box aspect ratios.

Localization Losses for Oriented Object Detection

Rotated Intersection over Union (RIoU) Loss^{ref.6}

$$RIoU = \frac{|B_{pred}^{rot} \cap B_{gt}^{rot}|}{|B_{pred}^{rot} \cup B_{gt}^{rot}|} \quad LOSS_{RIoU} = 1 - RIoU$$

Oriented Bounding Box OBB

Rotated Box $B(x, y, w, h, \theta)$



Often combined with Smooth L1 loss for precise parameter refinement (e.g., AO2-DETR^{ref.8}).



Angle periodicity issue
Gradient instability

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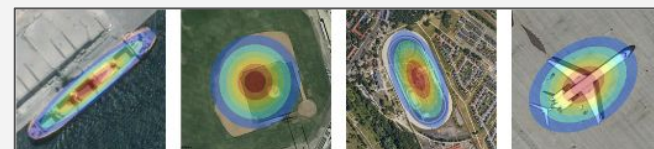
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Gaussian-based distance losses

- Based on the distance between 2D-Gaussian distributions.
 - Continuous & Differentiable with respect to the parameters
- **Convergence stability improvement during optimization.**

Gaussian Bounding Box GBB

2-D Gaussian Distribution $N(\mu, \Sigma)$



- Gaussian Wasserstein Distance (GWD) Loss^{ref.9}
- Kullback-Leibler Divergence (KLD) Loss^{ref.7}



Necessitate nonlinear mapping function
Involve additional hyperparameter tuning
Lack intuitive geometric interpretation

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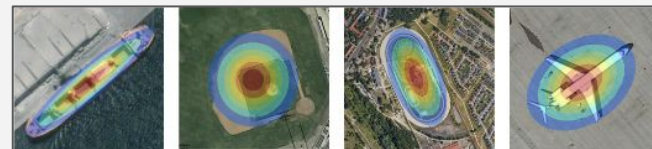
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Probabilistic IoU (ProbiIoU) Loss^{ref.10}

- Based on **Hellinger distance**

Potentials and limitations of ProbIoU Loss



ProbIoU Potentials

- 😊 ProbIoU = $1 - H_d$ correlates with the IoU values between elliptical regions of the GBBs, providing a highly intuitive geometric interpretation (1 for identical distributions and 0 otherwise).
- 😊 The resulting loss ($1 - \text{ProbIoU}$) is differentiable and satisfies all the properties of a distance metric.
- 😊 Does not require any nonlinear mapping or additional hyperparameters

Potentials and limitations of ProbIoU Loss



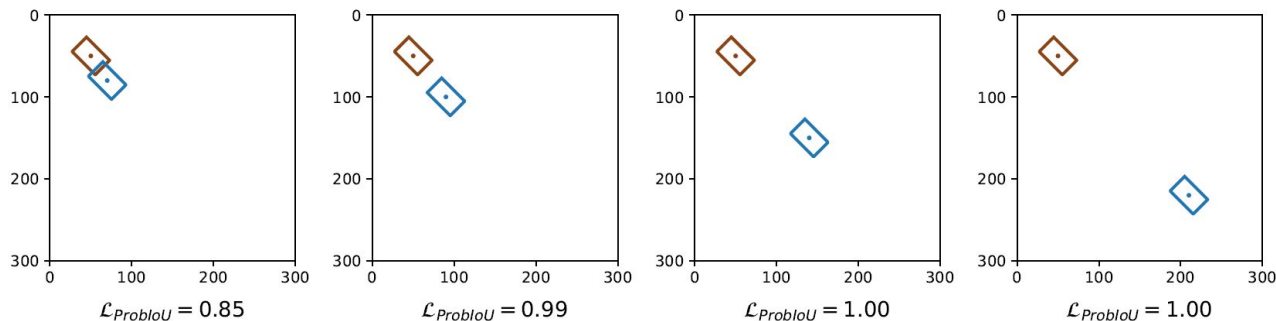
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ProbIoU Limitations



- 😞 Saturation of the Hellinger distance
 - No information about the relative proximity of the boxes
 - No guidance for optimization
 - Low discriminative power of the ProbIoU loss



Rethinking OBB-to-GBB Conversion (1/2)



OBB Representation

$$B(x, y, w, h, \theta)$$



GBB Representation

$$N(\mu, \Sigma)$$

ProbIoU

$$\mu_{ProbIoU} = (x, y)^T$$

$$\Sigma_{ProbIoU} = RD_{ProbIoU}R^T$$

$$= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \frac{w^2}{12} & 0 \\ 0 & \frac{h^2}{12} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

λ_w (red dashed circle around $\frac{w^2}{12}$)
 λ_h (green dashed circle around $\frac{h^2}{12}$)



KLD & GWD

$$\mu_{KLD/GWD} = (x, y)^T$$

$$\Sigma_{KLD/GWD} = RD_{KLD/GWD}R^T$$

$$= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

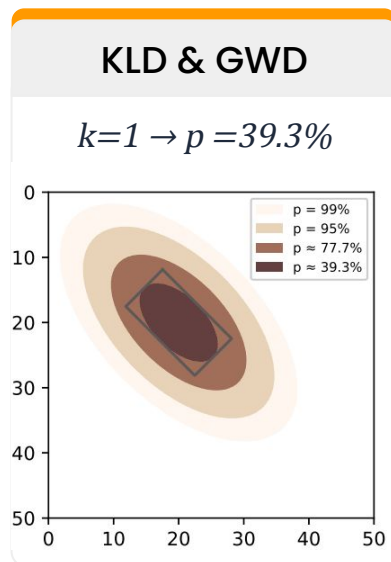
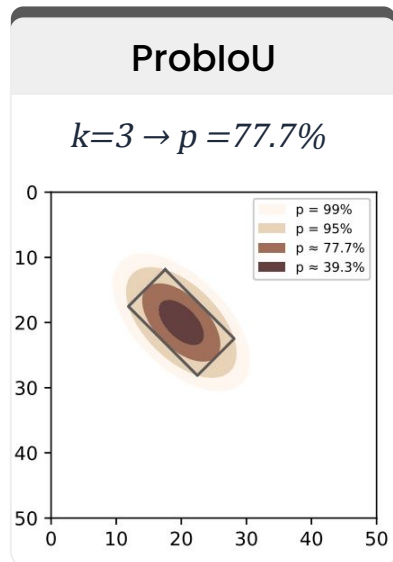
λ_w (red dashed circle around $\frac{w^2}{4}$)
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Rethinking OBB-to-GBB Conversion (2/2)

Semi-axis lengths of the largest ellipse inscribed in the OBB:

$$\frac{w}{2} = \sqrt{k\lambda_w} \quad ; \quad \frac{h}{2} = \sqrt{k\lambda_h} \quad ; \quad k = \chi_{2,p}^2$$



Generalized OBB-to-GBB conversion formula

$$\mu = (x, y)^T$$

$$\Sigma = RDR^T$$

$$= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \frac{w^2}{4k} & 0 \\ 0 & \frac{h^2}{4k} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

Proposed Method: ProbIoU+ Loss (1/3)

We propose to minimize the normalized distance between the means of the GBBs.

Let $GBB_1 \sim N(\mu_1, \Sigma_1)$ and $GBB_2 \sim N(\mu_2, \Sigma_2)$



$$\mathcal{L}_{ProbIoU^+} = \mathcal{L}_{ProbIoU} + \frac{\|\mu_1 - \mu_2\|_2^2}{s^2}$$

where:

- $\|\mu_1 - \mu_2\|_2^2$: the squared Euclidean distance between the mean vectors
- s : the normalizing factor



How to compute s using a Gaussian-based approach?

Proposed Method: ProbloU+ Loss (2/3)

STEP 1

Compute the mean and the covariance matrix of the Gaussian Mixture of GBB1 and GBB2

Mean

$$\begin{aligned}\mu_m &= \pi_1 \mu_1 + \pi_2 \mu_2 \\ &= \frac{1}{2}(\mu_1 + \mu_2)\end{aligned}$$

Covariance Matrix

$$\begin{aligned}\Sigma_m &= \pi_1(\Sigma_1 + (\mu_1 - \mu_m)(\mu_1 - \mu_m)^T) + \pi_2(\Sigma_2 + (\mu_2 - \mu_m)(\mu_2 - \mu_m)^T) \\ &= \frac{1}{2}\Sigma_1 + \frac{1}{2}\Sigma_2 + \frac{1}{4}(\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \\ &=: \begin{pmatrix} \alpha & \gamma \\ \gamma & \beta \end{pmatrix}\end{aligned}$$

Proposed Method: ProbloU+ Loss (2/3)

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STEP 2

Approximate the Gaussian mixture by the Gaussian distribution $N(\mu_m, \Sigma_m)$

aka “**Moment-Matched Gaussian Approximation**”

Proposed Method: ProbloU+ Loss (2/3)

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STEP 3

Set s to the major-axis length of the Gaussian's elliptical contour at confidence level p

$$s = 2\sqrt{k\psi_{\max}} \quad \text{where } k = \chi_{2,p}^2 \text{ and } \psi_{\max} \text{ is the maximum eigenvalue of } \Sigma_m$$

Proposed Method: ProbIoU+ Loss (3/3)



RESULTING EXPRESSION

$$\mathcal{L}_{ProbIoU^+} = \mathcal{L}_{ProbIoU} + \frac{\|\mu_1 - \mu_2\|_2^2}{s^2}$$



$$\mathcal{L}_{ProbIoU^+} = \mathcal{L}_{ProbIoU} + \frac{\|\mu_1 - \mu_2\|_2^2}{4k\psi_{\max}} \quad \text{where} \quad \psi_{\max} = \frac{(\alpha + \beta) + \sqrt{(\alpha - \beta)^2 + 4\gamma^2}}{2}$$



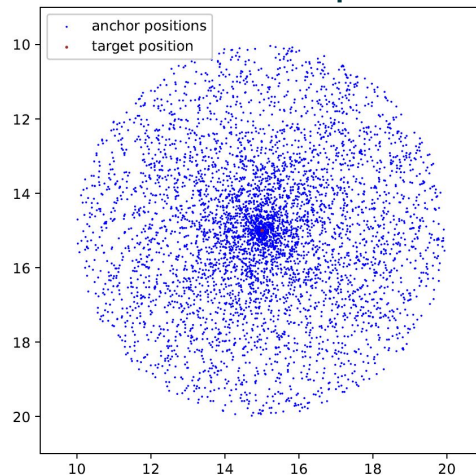
$$\mathcal{L}_{ProbIoU^+} = \mathcal{L}_{ProbIoU} + \frac{(x_1 - x_2)^2 + (y_1 - y_2)^2}{2k \left((\alpha + \beta) + \sqrt{(\alpha - \beta)^2 + 4\gamma^2} \right)}$$

Simulation Study

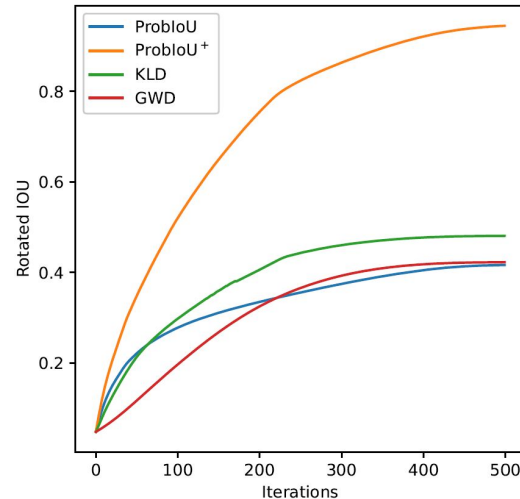
Table 1: Simulation parameters for target and anchor boxes.

Box type	Position	Area (scale)	Aspect ratio (w/h)	Angle (rad)
Target boxes	1 fixed position at the center of a circular region $\mathcal{D}((15, 15), 5)$	1 (unit area)	1, 2, 3, 4	$-\pi/2, -\pi/3, -\pi/6, 0, \pi/6, \pi/3$
Anchor boxes	5000 positions uniformly sampled within $\mathcal{D}$	0.5, 0.75, 1, 1.5, 2	1, 2, 3, 4	$-\pi/2, -\pi/3, -\pi/6, 0, \pi/6, \pi/3$

Initial Setup



Performance results



TOTAL
14.4M
Regression cases

Experiments (1/3)

Dataset	Task	Classes	Images	Train	Test
HRSC2016 ^{ref.11}	Ship Detection	Ships	1070	626	444

Table 2: Evaluation of different localization losses on HRSC2016 dataset

Dataset	Detector	Loss	AP ₅₀	AP ₆₀	AP ₇₅ ^{ref.12}	AP ₈₅	mAP
HRSC2016	RetinaNet	Smooth l_1	83.09	76.28	50.90	14.28	47.52
		GWD	87.82	82.84	55.56	14.61	51.40
		KLD	89.07	85.55	58.65	12.85	52.29
		ProbIoU	88.76	85.54	60.81	14.33	53.11
		ProbIoU ⁺	91.11	87.70	65.28	16.53	55.90
	R ³ Det	Smooth l_1	90.46	86.65	63.60	19.39	55.22
		GWD	94.06	90.46	62.87	16.12	55.86
		KLD	93.68	90.97	67.92	17.54	57.88
		ProbIoU	95.24	92.87	71.81	26.78	61.40
		ProbIoU ⁺	94.99	92.91	73.32	29.57	62.11


AP₇₅
+4.47%


AP₇₅
+1.51%

Experiments (2/3)

Dataset	Task	Classes	Images	Train	Test
ICDAR2015 ^{ref.13}	Scene Text Detection	Text	1500	1000	500

Table 3: Evaluation of different localization losses on ICDAR2015 dataset

Dataset	Detector	Loss	AP ₅₀	AP ₆₀	AP ₇₅	AP ₈₅	mAP
ICDAR2015	RetinaNet	Smooth l_1	71.04	62.76	32.52	5.85	35.78
		GWD	72.06	61.92	30.97	4.99	35.42
		KLD	71.25	61.34	28.34	3.79	33.98
		ProbIoU	71.73	62.13	31.72	5.23	35.47
		ProbIoU ⁺	72.75	64.61	32.91	5.07	36.42
	R ³ Det	Smooth l_1	73.26	63.53	28.81	3.28	35.16
		GWD	72.96	62.57	27.81	3.01	34.51
		KLD	73.66	62.05	25.59	2.67	33.94
		ProbIoU	73.72	63.59	29.38	3.49	35.53
		ProbIoU ⁺	73.70	64.74	31.68	4.00	36.05



AP₇₅

+1.19%



AP₇₅

+2.30%

Experiments (3/3)

Dataset	Task	Classes	Images	Train	Test
DOTA-v1.0 ^{ref.1}	Aerial Object Detection	15 Categories	2806	1869	937

Table 4: Evaluation of different localization losses on DOTA-v1.0 dataset

Detector	Loss	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
RetinaNet	Smooth l_1	88.74	73.68	42.49	67.80	77.26	58.93	76.40	90.88	76.95	82.60	56.78	63.01	53.73	63.53	44.01	67.79
	GWD	88.31	74.09	38.77	69.60	74.94	57.27	76.94	90.85	79.52	82.57	55.50	64.81	51.96	66.70	30.58	66.83
	KLD	88.95	75.97	39.75	70.17	74.99	59.91	77.23	90.85	76.40	82.66	52.26	61.69	52.06	65.64	38.56	67.14
	ProbIoU	88.64	75.86	39.28	69.73	78.01	62.92	77.65	90.86	80.77	82.56	53.83	60.49	56.98	67.29	35.48	68.02
	ProbIoU ⁺	88.84	77.64	42.90	66.72	78.25	63.32	77.83	90.86	79.83	83.41	55.65	65.90	59.60	68.01	47.75	69.77
R ³ Det	Smooth l_1	89.32	72.73	47.32	70.02	77.86	71.88	79.10	90.82	79.55	84.67	55.04	62.72	62.76	63.29	39.59	69.78
	GWD	89.46	73.69	42.40	70.52	75.46	70.59	79.15	90.86	79.80	83.92	54.97	62.03	54.97	67.05	48.28	69.54
	KLD	89.10	74.06	43.38	69.63	75.30	71.58	79.24	90.86	76.50	83.17	54.34	63.28	55.27	57.48	44.92	68.54
	ProbIoU	89.12	74.07	48.51	71.85	77.63	74.15	84.95	90.87	78.65	83.70	59.82	63.74	65.08	67.57	45.39	71.67
	ProbIoU ⁺	89.39	77.28	48.16	72.31	78.29	73.98	84.97	90.86	79.03	83.74	55.52	64.44	65.18	65.51	51.65	72.02



mAP

+1.75%



mAP

+0.35%

Top 2 performances are highlighted in red and blue, respectively

Conclusion



Main Contributions

01

Generalized OBB→GBB conversion for oriented object detection

02

ProbloU+ loss: a novel localization loss that enhances ProbloU using Gaussian distribution properties

03

Comprehensive evaluation: Data- and model-agnostic simulation & Experiments across public datasets

Thank You



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